

Internship Report

On

AI/ML Intern

Internship Duration

(01/11/2025) to (31/05/2026)

fx31 labs

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE

(with specialization in CSE)

Submitted By

DEVAM P SOLANKI
(Enroll. No. 20220702012)

under the guidance of

Ms. Anjali Joshi
Assistant Professor UIT, KU

UNITEDWORLD INSTITUTUE OF TECHNOLOGY

KARNAVATI UNIVERSITY

Uvarsad, Gandhinagar, Gujarat-382422

MAY 2026

INTERNSHIP COMPLETION CERTIFICATE

5/1/26, 12:11 PM

Karnavati University Mail - Internship Confirmation Status – Devam Solanki (20220702012) | Fx31Labs



Devam P. Solanki 20220702012 <20220702012@karnavatiuniversity.edu.in>

Internship Confirmation Status – Devam Solanki (20220702012) | Fx31Labs

1 message

Nirav Patel <nirav@fx31labs.com>

Thu, Apr 30, 2026 at 4:38 PM

To: Rama Maliya <rama@karnavatiuniversity.edu.in>

Cc: "Dr. Nilesh Modi" <nilesh@fx31labs.com>, "20220702012@karnavatiuniversity.edu.in" <20220702012@karnavatiuniversity.edu.in>

Date: 30th April 2026

Dear Faculty Member,

This email is to certify that **Mr. Devam Solanki** has been pursuing his internship program of approx. Seven months with **Fx31 Labs Private Limited**. His internship tenure is from 1st November, 2025 to 31st May, 2026. He has been working with Top management and Engineering Team, and has been actively & diligently involved in the projects and tasks assigned to him.

We truly appreciate his dedication, discipline, and eagerness to learn throughout the internship. His hard work and commitment have been commendable!

We wish him a bright future.

--

Thanks & Regards,

Nirav Patel

Growth Partner & Talent Head

+91 91047 54177

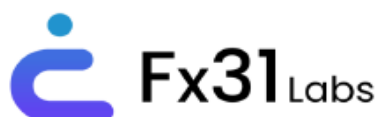
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INTERNSHIP OFFER LETTER

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Offer Letter

8th December, 2025

Devam Solanki,
Ahmedabad, India

Dear Devam,

On behalf of Fx31 Labs, we are pleased to welcome you with this offer letter for your volunteer Final Semester Project Internship through KU University with following terms:

Title/Position: Project Intern

Work Location: Hybrid (Fx31Labs, Ahmedabad)

Joining Date: 1st November, 2025

Duration: 1st November, 2025 to 31st May, 2026

We are an innovative and rapidly growing Rocketship. We know that you will be a valuable asset to our organization and will play an integral part in company's and world's future with your talent. We genuinely believe in your potential and talent. So in this role, you will be reporting directly to Ms. Disha Sharma (Solution Architect).

During your employment with Fx31 Labs, you may have access to trade secrets and confidential or proprietary business information belonging to the company and/or the clients. By accepting this offer, you acknowledge that this information must remain confidential and agree to refrain from using it for your own purposes or disclosing it to anyone outside of Fx31 Labs.

This offer letter represents the full extent of the offer and supersedes any prior conversations about the position. Changes to this agreement may only be made in writing. If you have any questions about this offer, feel free to contact us. Kindly review this offer letter in full, and sign it using ZohoSign to confirm your acceptance of the position no later than 7pm on December 8, 2025.

You agree that during the period leading to your employment, in case any asset such as laptop/mobile and/or any gadget are issued in your name by Fx31 Labs, the same shall always be the property of the company

+19172164038 (USA) 
+919825721363 (India)

contact@fx31labs.com 

225 Hartshorn Drive, Short Hills, NJ 07078 (USA) 
205, Amrakunj Avis, Chandkheda, Ahmedabad, Guj 382424 (India)



and you under no circumstances shall create/ claim any rights whatsoever. This clause shall be applicable during the term of the employment as well as upon.

We look forward to having you begin your professional journey at Fx31 Labs and wish you a successful and great future with us. Excited to have you on our team!

Best Wishes,

For "EMPLOYER"

Nirav Patel

Authorized Signatory

I, Devam Solanki, accept the above offer.



Devam Solanki,
8th December, 2025

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+919825721363 (India)

contact@fx31labs.com 

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NOC from CSD KU



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www.karnavatiuniversity.edu.in

Date: 1/27/2026

To Whom It May concern

This is to certify that Devam Solanki bearing enrolment No. 20220702012 is a bonafide student of Unitedworld Institute of Technology (UIT), Karnavati University, currently enrolled in the 8th Semester of B.Tech- CSE, for the Academic Year 2025-26.

The student has been approved by the Career Services Department (CSD) to pursue an internship at Fx31 Labs.

The internship will be considered as part of the student's mandatory academic requirement, subject to successful completion and submission of the following to CSD.

- Internship Offer Letter
- Internship Completion Certificate

This approval is issued for official and academic purposes.

For any further information, please feel free to contact the undersigned.

Authorized Signatory

Shayan Pal

Director, Career Services

Karnavati University | [UID](#) | [UIT](#) | [UIM](#) | [UWSL](#)

UID | UIM | UWSL | USLM | UIT | KSD | KSR

DECLARATION BY THE STUDENT

I, **Devam Solanki, 20220702012**, student of Bachelor of Technology in CSE/CSE-AIML hereby declare that the internship report titled “**AI/ML Intern**” which is submitted by me to Unitedworld Institute of Technology, Karnavati University, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science (with specialization in CSE) has not previously formed the basis for the award of any degree, diploma, or other qualification.

I further declare that due permission has been obtained for the use of any copyrighted material included in this internship report, except for brief excerpts that require only proper acknowledgment in scholarly writing, and all such use has been appropriately acknowledged.

Gandhinagar

26/04/2026

Devam Solanki
20220702012
B. Tech CSE

KARNAVATI UNIVERSITY
Uvarsad, Gandhinagar, Gujarat-382422
Unitedworld Institute of Technology

BONAFIDE CERTIFICATE

This is to certify that **Devam Solanki, 20220702012** a student of B. Tech CSE/CSE-AIML, has successfully completed the internship report titled “**Project Intern**”, submitted to Unitedworld Institute of Technology, Karnavati University, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology.

This report is a bona fide record of the work carried out by the student under my guidance and supervision. To the best of my knowledge, this work is an original contribution and has not been submitted, either in part or in full, for the award of any degree, diploma, or any other qualification at this University or elsewhere.

Date: 26/04/2026

Place: Gandhinagar

Ms. Anjali Joshi
Assistant Professor
Unitedworld Institute of Technology

(External Examiner)

Dr. Rebakah Geddham
Assistant Dean
Unitedworld Institute of Technology

Dr. Puneet Sharma
Associate Dean

Unitedworld Institute of Technology

ACKNOWLEDGEMENT

I would like to convey my heartiest thanks to my organization guide **Mr. Nilesh Modi, fx31 labs, Gandhinagar, Gujarat**, for their invaluable guidance and continuous support throughout my internship. Their expertise, problem-solving approach, and encouragement have been instrumental in enhancing my learning experience.

I especially thank my guide **Ms. Anjali Joshi**, Assistant professor, Unitedworld Institute of Technology, Karnavati University for all his guidance and support. His extreme patience has guided me in situation of need for which I am extremely grateful.

I wish to express my sincere thanks and deep sense of gratitude to **Dr. Puneet Sharma**, Associate Dean, UIT, and **Dr. Rebakah Geddiam**, Assistant Dean, UIT, Karnavati University, for their valuable support and encouragement.

I am also thankful to our coordinator, Ms. Rama M. Maliya and UIT Placement Team for their continuous guidance.

I would like to acknowledge the Career Service Department (CSD), Karnavati University for their support in fulfilling my internship requirements. Their timely assistance played a crucial role in the successful completion of my work.

I am thankful to my friends for their unwavering support, encouragement, and willingness to help me whenever needed.

Above all I am indebted to my beloved parents, whose blessing and best wishes have gone a long way in making this semester internship a grand success.

Devam Solanki
20220702012
B. Tech CSE

ABSTRACT

AutoSim addresses the hallucination problem faced by large language models when solving differential equations through an architectural method rather than prompting. It does so by distinguishing between natural language processing and computation through its three-tiered architecture, consisting of a Next.js frontend for data input and visualization, and a FastAPI backend with six deterministic methods, including symbolic, Laplace, and numerical methods like Runge-Kutta and Euler, all of which rely on solid libraries like SymPy, SciPy, and NumPy for computation.

One of the biggest advantages of AutoSim is that it has a verification strategy of six sources: analytical comparison verification, external tool verification, physical inspection verification, method verification, residual verification, and initial condition verification. All these verification techniques produce a confidence level for the results obtained, and the results are reliable for different types of equations.

The language model is only used for interpretation and explanation generation; it is not used for any calculations, which prevents hallucinations by design. AutoSim provides a reliable, informative, and production-ready solution for solving ordinary differential equations with features such as comparison of different methods and strong test results.

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Chapter 1

Introduction

1.1 Introduction to the Internship

The innovations that aim to ensure success for organizations in the rapidly evolving technological landscape by means of increasing productivity, transforming processes, and achieving strategies are developed by Fx31 Labs, an industry leader in the field of technology with experience in sophisticated technologies and Generative AI (Gen AI). The internship was conducted from November 1, 2025, to April 30, 2026, in Fx31 Labs, an industry leader in technology research with specialization in artificial intelligence and computational science.

The creation and development of AutoSim, a commercial-grade Ordinary Differential Equations (ODE) solver software that utilizes a zero hallucination framework, have been the primary focus of my work for this internship. The absence of a free and accessible computational tool that uses multiple methods to solve differential equations, independently validate each solution with at least six sources, and provide a detailed step-by-step derivation in an understandable way for students and experts alike is a significant void in scientific software development that AutoSim addresses [2][9]

The system is built using Next.js 14 with Tailwind CSS 4 for the production frontend, which utilizes a dark scientific theme along with interactive Canvas 2D visualizations, Python 3.11 for the backend computation layer (only making use of SymPy, SciPy, and NumPy for any mathematical calculations), and FastAPI for the RESTful API layer. One of the research efforts carried out during the course of this internship included an analysis of physics informed neural networks, universal differential equations, program-assisted language models, and chain-of-thought prompting [1][3][4][5][7][8].

1.2 Objectives of the Internship

There were six key goals relating to technology and professionalism that structured this internship, and each goal corresponded to an important element of the AutoSim system. In total, these goals describe the expected skillset or achievements gained throughout the project.

1. Understanding the Full Process of an ODE Solver gained experience in building modular structures that have multiple components within scientific software systems.
2. ODE Solver Execution and Performance Analysis developed competence in numeric and symbolic calculations by practicing approaches like Symbolic solutions using SymPy, Laplace Transform, RK45, Radau, Forward Euler, and Improved Euler.
3. Design of an Architecture with Zero-Hallucination established a system in which all calculations are done through trusted mathematical libraries and the only job LLMs are assigned is interpreting
4. Development of a System for Multiple Source Checks gained experience in cross-verifying calculated results using six different sources in order to verify answers physically and mathematically.

1.3 Scope of the Work

The AutoSim system has six functional areas that set out the boundaries of this internship. Table 1.2 shows what Phase I of this project includes and what will be excluded from it and handled at a later phase.

1. Scope of Router and ODE Parser: The rules for selecting the solver, detecting the order, linearising the equations, and normalisation of the equations have been included. Systems of ODEs and PDEs have not been considered. Only scalar ODEs with one variable $x(t)$ have been considered.
2. Six Solver Methods Covered: Some of the solver methods, including the Symbolic, Laplace Transform, RK45 (Runge-Kutta-Fehlberg), Radau, Forward Euler, and Improved Euler, were implemented and verified. Not within Scope: Solutions to ODEs with variable coefficients using higher-order Laplace Transforms, as well as solutions to systems >2 , are excluded from the scope of this project.

3. The six-step validation approach includes analytical library validation, Wolfram Alpha validation, energy conservation law, consistency between various approaches, residual error, and IC fulfillment. Out of Scope: If there is no API key available, the validation using Wolfram Alpha could be skipped since it requires an API key.
4. Next.js Frontend In Scope: With the use of MathJax, I developed a dark-mode-based Single Page Application (SPA), which consists of 2D canvas visuals, a result explorer with four tabs, solver selection, and derivation using LaTeX. Out of Scope: Offline capabilities and mobile apps are deferred to future phases.
5. FastAPI Successfully designed JSON serialization mechanism for numbers (such as NumPy arrays) and REST API interfaces such as /solve, /verify, and /llm/interpret. Phase I will not include cloud-based deployment, user management, and authentication mechanisms.
6. The integration of an optional API feature allows for natural language interpretation of the calculated values. The third phase will see tuning and hosting of the model; LLMs will be limited exclusively to interpreting calculations and won't perform any computation.

Chapter 2

Company Profile

2.1 Introduction to the Company

Fractal31 (Fx31 Labs) is an innovative company providing software development services as well as doing advanced technology research and development. This company has two major centers, one being headquartered in Short Hills, New Jersey, USA and another is based in Chandkheda, Ahmedabad, Gujarat, India. The company has been founded by Dr. Nilesh Modi, who has a total experience of over 25 years and a Ph.D. in Electrical & Computer Engineering from the University of California. He has earlier worked for leading Fortune 100 companies such as Intel, Cadence, IBM, and Cygnet Infotech.

Fx31 Labs operates at the convergence point between enterprise-level software development, deep tech applications, and artificial intelligence. Providing Fortune 500-grade product development services to start-ups, growth-stage enterprises, and firms at a substantially lower price point is what distinguishes this organization. This is mostly accomplished through a blend of remote engineering and on-site partnership model. Fx31 Labs is affiliated with IIMA Ventures, which represents the venture capital arm of the Indian Institute of Management Ahmedabad. Forbes India has featured this organization among its list of Select 200 companies. Moreover, GoodFirms has ranked it as one of the best software development companies.

Considering that the firm's talent retention ratio is 97% and on time deliveries ratio is 95%, the company, which mainly works in two regions; the United States and India, has been able to complete more than 100 projects. The positioning of Fx31 Labs, according to the firm, lies in offering access to the 3% best engineers who combine effective communication skills with technical expertise since, according to the firm, lack of such often leads to failure in offshore software development partnerships.

2.2 Vision, Mission and Core Values

Vision

Vision of Fx31 Labs is that companies ranging from the Fortune 500 firms to young startups will be able to harness cutting-edge artificial intelligence and advanced technology, which would help them develop innovative solutions for some pressing problems around the world.

Mission

The objective at Fx31 Labs is to realize the full power of applied deep technology and generative AI through assembling world-class engineering teams, developing MVPs rapidly, and providing strategic AI consultancy, which can help clients bring their innovative ideas to the market and transform them into production-grade products. Clients who require not only the services of the engineering team but also technological strategy and vision can also benefit from the company's remote CTO offer.

Core Values

These values underpin the workings of Fx31 Labs:

- **Technical Excellence:** The company retains its technical employees at a high rate of 97%, while its pool of talent comes from among the top 3% of candidates that pass pre-screening.
- **Fast Innovation:** The company's focus on MVPs, fast prototyping, and an average of two days for its initial candidate interview suggests a preference for quick execution.
- **Client Partnership:** From what one may learn from the testimonials posted on their site, the company places importance on client partnership with a particular emphasis on flexibility and adaptability in business models.
- **Integrity and Confidentiality:** The offer letter presented to both interns and regular employees ensures the absolute integrity towards the confidential nature of the client's business secrets and knowledge.

- **Global Inclusivity:** An inclusive approach with regard to the global distribution of talents for ensuring quality yet affordable product development through recruitment of engineers from India and serving the clients in the USA and India.

2.3 Products and Services Offered

Fx31 Labs offers end-to-end AI-enabled technology services such as strategic consulting, product development, and talent augmentation. Here is a description of the main service lines.

AI-Powered Deep Tech Talent Enhancement

Providing pre-screened, deep-technology capable engineering manpower on a staff augmentation basis is Fx31 Labs' flagship service. In order to support or oversee technical workstreams, clients integrate Fx31 Labs personnel directly into their teams. Compared to comparable personnel acquired domestically in the US or Europe, the organization claims a 30–50% cost advantage and a 100% hiring success rate.

Development of MVPs through Generative AI

For the next generation of entrepreneurs and startups, the MVP and GenAI Product Lab of Fx31 Labs is available to assist them in developing their MVPs using Generative AI techniques along with efficient product engineering experts. This business can help resolve problems like costly software development and extended time to market experienced by many startups.

Generative AI and Machine Learning Consulting Services

It offers generative AI and machine learning consulting services in which model selection, solution architecture, tuning, implementation, and optimization of the service are covered. Companies interested in implementing AI agents and LLM systems within their existing business models will find this service useful.

Fx31 Labs: Coderbotics AI

This exclusive platform by Fx31 Labs uses Artificial Intelligence to aid code writing, review, and validation. The AI technology will help make the software development process faster without compromising on quality. This platform represents the investment made by the company in AI technologies that aid in delivering its engineering solutions.

Additional Services Lines

In addition to the above, Fx31 Labs offers software outsourcing, remote CTO advisory services, hiring an engineering team, services related to mobile app and web development in a variety of technologies, and new technology development, such as those related to robotics, blockchain, Internet of Things, and Augmented Reality/ Virtual Reality (AR/ VR). The technology stack of the organization comprises more than forty technologies ranging from front-end, back-end, databases, cloud, and DevOps, to AI/data science, and others.

Table 2.1. Summary of Fx31 Labs Service Portfolio

Service Line	Description
AI Talent Augmentation:	Through a business-oriented yet flexible approach, pre-screened deep-tech engineers are incorporated into the team of clients
Development of MVP	Quick delivery of MVP and prototypes for innovative startups and new businesses using GenAI
LLM-based Generative AI/ML	LLM implementation, deployment of models, full-fledged AI strategy, and fine-tuning services
Coderbotics AI	An innovative solution for fast coding and development using artificial intelligence
Mobile & Web Development	Development of full-stack apps using technologies like React, Next.js, Flutter, Node.js, and Python and more for web and mobile

Remote CTO Services	Strategic guidance from remote CTO for organizations without an in-house CTO
Software Outsourcing	Full-scale software development using an offshore development model with project managers assigned to projects
Dedicated Teams	Permanent dedicated engineering teams working according to client product roadmaps
Robotics & IOT	The latest trends in robotics, IOT, and AR-VR

2.4 Structure of Organisations

Being a growth tech company, Fx31 Labs follows a lean leader model in its operation. This includes a strong senior leadership team and functional leaders in engineering, artificial intelligence, product, and customer solutions.

Team Leadership

The strategic direction for the organisation's AI products and research is determined by its founder and CEO, Dr. Nilesh Modi. Kushagra Saxena serves as an investor and board advisor to the organisation. He has been working in the capacities of Global CTO at MasterCard, CEO of PagoNxt Merchant Solutions, and McKinsey. This provides strategic direction for the organisation in terms of working with the enterprises and financial sectors.

Functional Structure Relevant to the Internship

Fx31 Labs internship program was based on the concept of Applied Scientific Computing and AI Research Practice. Mr. Disha Sharma, who works as a Solution Architect, was the immediate supervisor and technical consultant of the intern. This functional structure enabled the intern to be involved in applied scientific computing and full-stack engineering as well as design of AI systems by working directly under the supervision of an experienced professional.

Co-location and distributed teams were two types of teams in which the intern got exposure due to the hybrid work environment practiced by Fx31 Labs. The internship program involved on-site work at Ahmedabad premises and remote software development.

Chapter 3

Work Profile and Responsibilities

3.1 Internship Role and Designation

1. The Fx31 Labs Internship Program was designed for seven months, starting from November 1, 2025, and ending by May 31, 2026. As per the letter of offer issued by Fx31 Labs on December 8, 2025, I have been assigned as a project intern based on my designation.
2. Following the standard protocol of general engineering practices followed by Fx31 Labs, a mixed approach was used for work purposes including virtual as well as physical work in the Ahmedabad Fx31 Labs office.
3. With the aim of meeting certain requirements for B.Tech. Degree offered by Unitedworld Institute of Technology Karnavati University, the voluntary internship project was undertaken.
4. Dr. Nilesh Modi who is the CEO and founder of Fx31 Labs served as the immediate supervisor to help guide me in the internship process by assisting in architectural as well as design-related matters.
5. The final output product generated after the completion of the internship program was AutoSim, which was developed using an engineering perspective by following a zero-hallucination system approach towards ODE solving systems.

3.2 Tasks Assigned

The tasks for this internship spanned four development phases during a seven-month period from November 2025 to May 2026. There were specific deliverables, criteria for acceptance, and test gates to pass before moving onto the next phase. Table below provides a breakdown of tasks carried out on a weekly basis throughout the period of 23 weeks, followed by descriptions of the various phases.

Table 1: Week-by-Week Task Breakdown

Week	Phase	Description
Week 1 (01/11/2025 to 07/11/2025)	Phase 1	Responsibility held for the onboarding process that entailed setting up of the development environment, induction process by the supervisor and briefing on the AutoSim project.
Week 2 (08/11/2025 to 14/11/2025)	Phase 1	Performed research pertaining to the literature survey of hallucinations in large language models pertaining to the mathematical reasoning capabilities of the model as well as architectures such as Neural ODEs and PINNs [3][4].
Week 3 (15/11/2025 to 21/11/2025)	Phase 1	Carried out research related to the areas of chain-of-thought prompting, parameter-efficient techniques (LoRA & QLoRA), universal differential equations, and program aware language models and reported on the same for submission to the supervisor.
Week 4 (22/11/2025 to 30/11/2025)	Phase 1	The Software Requirements Specification (SRS) is finalized; scoping of the project is done by making the list of deliverables for Phase I and planning the schedule of deliverables for Phases II and III; and the decoupled three-tier architecture based on Next.js, FastAPI, and Python solver modules is designed.
Week 5 (01/12/2025 to 07/12/2025)	Phase 2	LHS/RHS splitting and order detection capability have been implemented; the ODEComponents class with unit tests has been added; and the ODEParser module using the normalized notation via regular expressions is built.
Week 6 (08/12/2025 to 14/12/2025)	Phase 2	Routes for symbolic solving, stiff equations, and some nonlinear equations, including the Duffing and Van der Pol equations, have been determined; ODEParser functionality

Week	Phase	Description
14/12/2025)		has been improved to extract the coefficients and linearity of the equation; and ODERouter using the deterministic decision tree method has been developed.
Week 7 (15/12/2025 to 21/12/2025)	Phase 2	Implemented the DerivationStep class for LaTeX formatted steps that include initial condition calculations, created the SymbolicSolver class by utilizing dsolve() function of SymPy library, and derived the characteristic equation with root classification and solution formation.
Week 8 (22/12/2025 to 28/12/2025)	Phase 2	Finalized the characteristic equation with root classification and solution formation, wrote unit tests for validating the solver properly, detailed alternative methods using the RoutingDecision class, and added the concept of a two-level fallback strategy for the SymbolicSolver class from the undetermined coefficient technique to the RK45 method.
Week 9 and 10 (29/12/2025 to 11/01/2026)	Phase 2	In this phase we have converted higher order into first-order system and developed NumericalSolver class with RK45 based method using SciPy solve_ivp wrapper and then we have verified it through numerical solver using unit testing. Additionally, using SciPy's RK45 approach for stiff type of equation we have implemented another solver which uses Radau. The solution is validated using end to end tests.
Week 11 (12/01/2026 to 18/01/2026)	Phase 2	With the help of supervisor from fx31labs we have reviewed all the code and fixed some of the issues. We also documented the same to get the rough idea about the problems we have solved and our next steps.

Week	Phase	Description
Week 12 (01/02/2026 to 07/02/2026)	Phase 3	In this week another phase got started and we have looked it into solving ODEs by using forward Euler algorithm and developed EulerSolver based on the same. It uses a constant step integrator.
Week 13 and 14 (08/02/2026 to 21/02/2026)	Phase 3	In this week we have developed another Solver called LaplaceSolver and it was created using a six-step transform derivation algorithm.
Week 15 (22/02/2026 to 28/02/2026)	Phase 3	This week we have focused on making solution verification system by incorporating numerical differentiation. ODEVerifier class was developed with an analytical library of ODEs.
Week 16 (01/03/2026 to 07/03/2026)	Phase 3	During this time period we have test various initial conditions along with some threshold values and errors. Along with this cross-method verification WolframAlpha API were integrated to the ODEVerifier class.
Week 17 (08/03/2026 to 14/03/2026)	Phase 3	We have done some research work here were we found some specific techniques-based criteria for Euler methods to decrease false positives, and created VerificationReport class which contains rating and overall rating calculations.
Week 18 (15/03/2026 to 21/03/2026)	Phase 3	Developed serializer for NumPy arrays, complex numbers, SymPy expressions, did request validation based on Pydantic, described JSON response format and created FastAPI service with several REST APIs.

Week	Phase	Description
Week 19 (22/03/2026 to 28/03/2026)	Phase 3	Performed automatic documentation validation in the OpenAPI format, finished testing of all APIs via Postman, launched Uvicorn ASGI server with CORS enabled for the Next.js frontend, and finished Phase 3 with code review and merging of branches upon successful completion of all tests.
Week 20 (01/04/2026 to 07/04/2026)	Phase 4	As the next phase begins with phase 4 we created a single page application with dark theme with the use of tailwind CS, implemented client-side data which fetched using "use client" along with some useEffect, and built the Solver Workspace input form for the equations.
Week 21 (08/04/2026 to 14/04/2026)	Phase 4	Visualization is an important part while solving the equation to we have created a component using Canvas API with an interactive phase portrait with toggle options and a time series graph.
Week 22 (15/04/2026 to 21/04/2026)	Phase 4	As for LaTeX equations MathJax 3 was put on the webpage for rendering. Additionally, Solution exploration pane with 4 different panes like Solution, Steps, Verification and LLM Insights), with confidence level banner are added.
Week 23 (22/04/2026 to 28/04/2026)	Phase 4	The "Compare All" feature allowing multiple solvers to run simultaneously and their results to be displayed at once was added, included the LLM Insights pane powered by the Anthropic API, and made sure that all outputs generated by the AI were different from the output layer of the computation.

3.3 Acquired technical and management skills

During the internship in Fx31 Labs, great progress was made in both categories: technical and professional competency. The skills acquired are categorised below.

Technical Skills:

1. **Backend Development using Python:** Through ten modules in backend development, the intern developed proficiency in the use of Python 3.11 for production-level work. Some of the areas covered include architectural design for software using modularity, creation of models through dataclasses, matching using regular expressions, OOP programming along with good definition of interfaces, handling errors and fallbacks, and serialization of complex python objects to JSON format.
2. **Scientific Computing Libraries:** The internship experience has provided me extensive exposure on how SciPy (solve_ivp() function that is used for numerical solutions to differential equations using RK45, Radau, BDF, and LSODA methods), NumPy (arrays, gradients, interpolation), and SymPy (symbolic manipulation and solution of ODEs via dsolve(), solution of equations using Laplace and Inverse Laplace transform, partial fractions). These three essential libraries form the Python suite of scientific computing.
3. **Full Stack Web Development:** The process of building the AutoSim front-end and back-end helped the intern gain proficiency in full stack web development. Some of the skills gained on the front-end include Next.js 14 (server-side rendering, routing, hooks, one-way data binding), TypeScript, Tailwind CSS 4, 2D API for interactive science graphics, and MathJax for rendering mathematical formulas using LaTeX notation. Some of the skills gained on the back-end include designing a web application with FastAPI, data validation using Pydantic, asynchronous programming in Python with Uvicorn, RESTful API design, and OpenAPI specifications.
4. **Testing Approach for Software:** The unit testing was conducted using pytest and the unittest.mock library, integration testing used pipelining testing with actual

dataclasses objects, system testing was done manually across different browsers using Postman for API routes, and validation testing compared solver results with analytical solutions using relative error calculations.

Professional and Managerial Skills:

1. **Project Planning and Phasing Skills:** The skills of task estimation, milestone identification, and phased delivery were developed by following the internship schedule which spanned for a period of seven months and had four phases of development. The structured approach allowed direct involvement with the management of the SDLC process as each new module was created in isolation using a feature branch of Git and only after successful testing.
2. **Technical Documentation:** Apart from the existing internship report, the internship involves producing a detailed nine-chapter capstone report. The ability to communicate intricate concepts in both mathematics and technology in writing was acquired through this process. This is directly useful in the field when working as a software engineer, in artificial intelligence research, and as a technical product manager.
3. **Version Control:** Through the utilization of GitHub and Git in the day-to-day development of the project, employees were exposed to how version control is done in the industry.
4. **Communication and Supervision:** The skill of effectively communicating one's technical work to a senior professional, supporting one's design rationale using proof, and integrating feedback constructively in the design process was attained by engaging in frequent technical talks about architecture decisions, design evaluations, and test results with Dr. Nilesh Modi (CEO & Founder).

Chapter 4

Project/Work Done

4.1 Description of the Work

AutoSim, an industrial-strength Ordinary Differential Equations (ODE) solver framework utilizing a zero hallucination approach, was the key project achieved through this internship. The AutoSim solver employs six distinct approaches to solve the differential equations, verifies each result from six sources, and shows the results using a modern, dark-themed dashboard accessible via any web browser without requiring installation.

AutoSim addresses three problems inherent in the existing ODE tools: the pedagogic problem (numerical solvers provide the result but not the mathematical process), the validation problem (traditional solvers provide one result without independent validation), and the access problem (there is no free, browser-accessible platform for solving and comparing ODEs using multiple approaches).

The zero hallucination architecture acts as the underlying architecture, where SymPy, SciPy, and NumPy conduct all the calculations. Anthropic API does not perform any mathematical calculations; instead, it may be optionally employed to interpret precomputed results in natural language. The full pipeline diagram is presented in Figure 4.1[2][9]

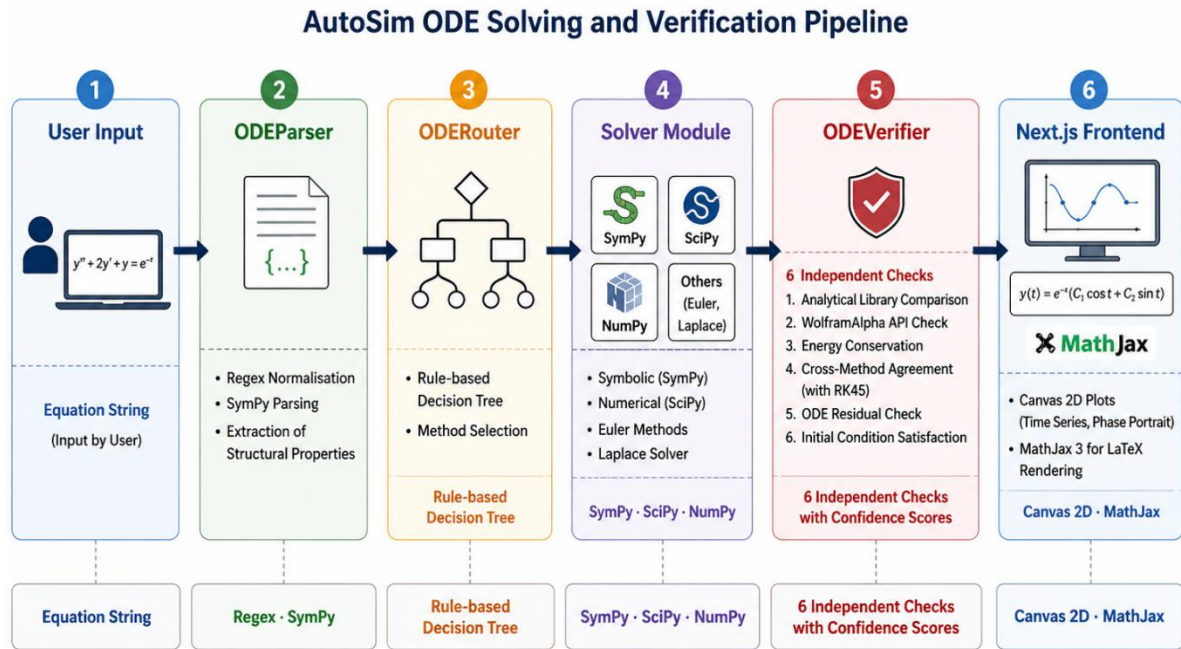


Figure 4.1. AutoSim System Pipeline

The key structural aspect of the design of this project is the strict segregation of the calculation and analysis stages, illustrated in Figure 4.2.

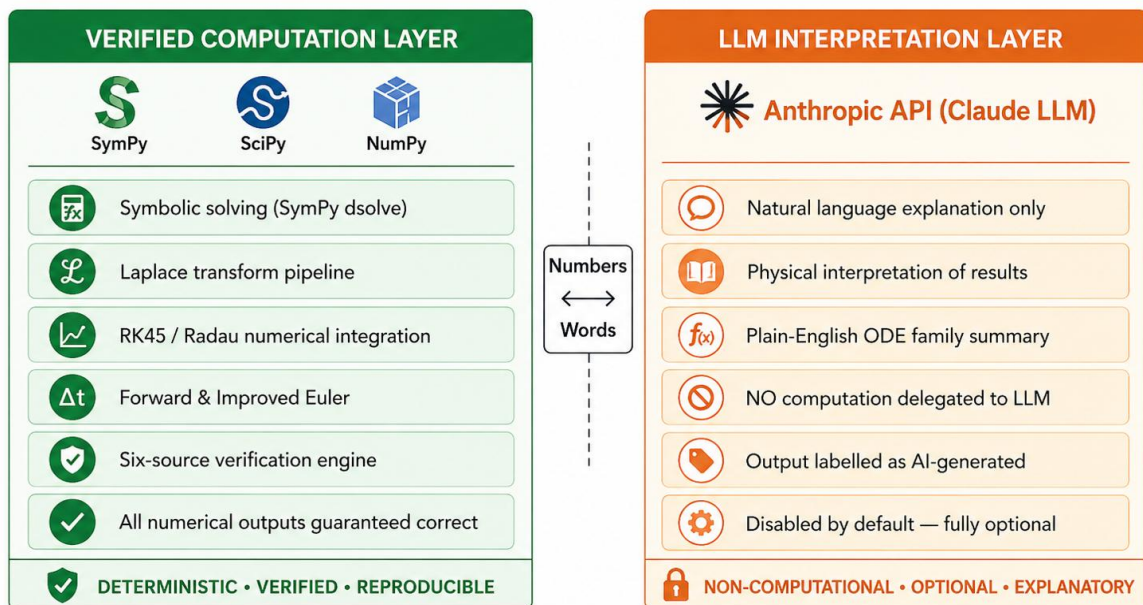


Figure 4.2. Zero-Hallucination Architecture Computation vs Interpretation Boundary

Figure 4.3 shows the six solving methods, along with their type classification and descriptions. The Compare All option runs all applicable algorithms simultaneously and displays the output on one Canvas graph for comparative purposes.

Characteristic	Type	Method
Symbolic (SymPy)	Exact	Characteristic equation, root classification, closed-form solution, apply ICs
Laplace Transform	Exact	$L\{ODE\} \rightarrow X(s)$, partial fractions via <code>apart()</code> , inverse $L \rightarrow x(t)$
RK45 (Dormand–Prince)	Numerical	Adaptive RK 4th/5th, step control (<code>rtol=1e-8</code>), best for smooth problems
Radau (Implicit RK5)	Numerical	Implicit RK (order 5), Newton iteration/step, handles stiff systems (ratio > 100)
Forward Euler	Educational	$y(n+1)=y(n)+h \cdot f(t,y)$, global error $O(h)$, first 10 steps stored
Improved Euler (Heun)	Educational	Predictor–Corrector, global error $O(h^2)$, step table displayed

Figure 4.3. AutoSim — Six Solving Methods with Type Classification

Upon calculating the answers, the automatic verification tool, as outlined in the following points, conducts six different tests. Each source produces its own percentage of confidence, whether the test passes or fails, and a number measuring the error; the overall confidence level is computed by taking the arithmetic average of all active sources.

1. **Known Analytical Library:** For comparison with preselected ODE family and an analytical solution, a relative error of less than 5% needs to be achieved.
2. **WolframAlpha API:** The WolframAlpha API checks WolframAlpha at sample points in time, but only if the API connection is available, using a relative error requirement of less than 5%.
3. **Cross-Methods Agreement:** This will ensure the relative discrepancy is less than 5% by separately solving the differential equation via RK45 and taking the difference from multiple points.
4. **Residual Error for ODE:** This is done to confirm that the residual error is less than 5% of the solution by using numerical derivatives to plug the obtained solution back into the original equation.
5. **Satisfaction of Initial Condition:** It will verify that the solution satisfies the initial condition with a tolerance of 1×10^{-2} at time zero.

The ODERouter decision logic, which determines which solver is selected for any given equation, is shown as a structured decision tree in Figure 4.4. Every routing decision is fully deterministic and traceable to a documented rule.

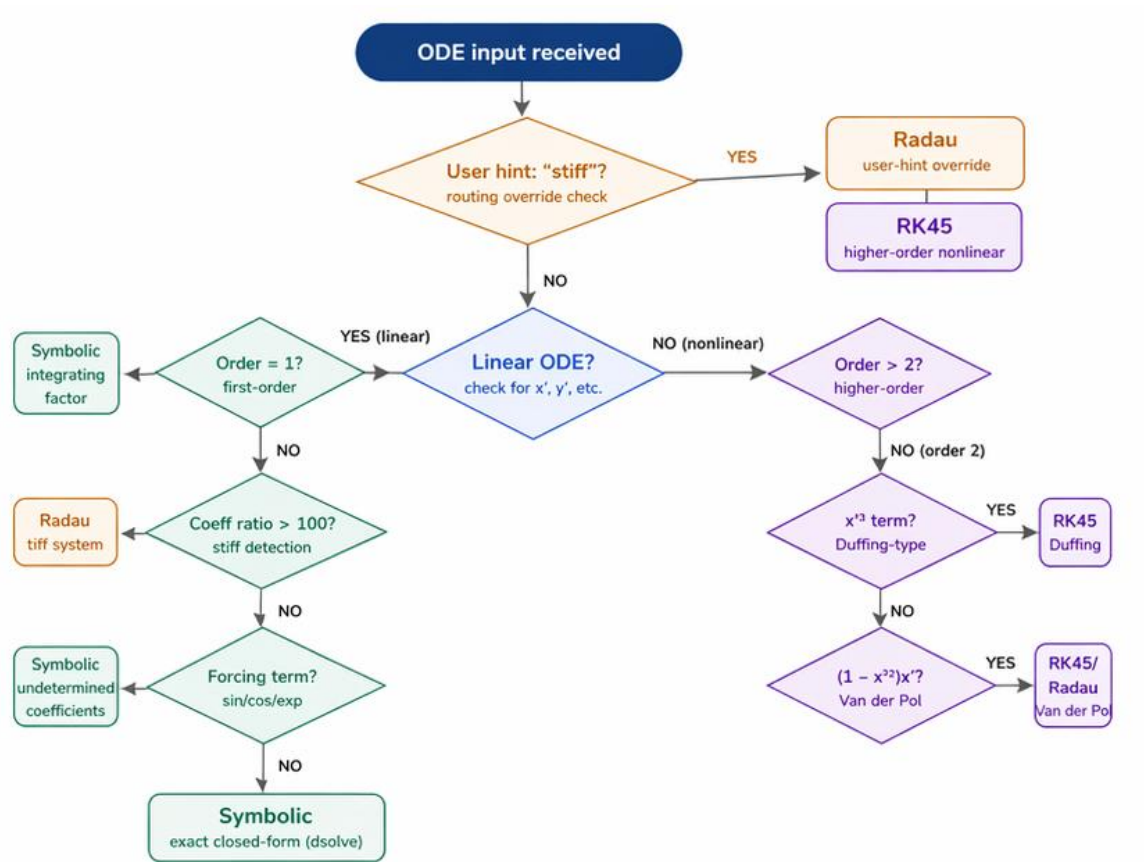


Figure 4.4 ODERouter Decision Flowchart

4.2 Technologies/Tools Used

The technology stack consists of three fully open source tiers. The technology stack is illustrated in Table 4.1 with the corresponding usage cases.

Table 4.2. Tools and Technologies Used

Layer	Tool / Library	Category	Role in AutoSim
BACKEND	Python 3.11	Backend Language	all calculations are performed within the solver pipeline, API and verification engine
	SymPy 1.12	Symbolic Maths	dsolve(), laplace_transform(), inverse_laplace_transform(), apart(), lambdify()

Layer	Tool / Library	Category	Role in AutoSim
	SciPy 1.11	Numerical calculations	find_peaks() to calculate the period; solve_ivp() to implement RK45/Radau/BDF
	NumPy 1.26	Array Operations	Array operations consist of statistics functions, gradients for residuals validation, and arrays of solutions
	FastAPI 0.104	Backend API	automatic generation of OpenAPI docs and RESTful API with Pydantic validation
	Uvicorn 0.24	ASGI Server	High performance async server with hot reload at development time
	Pytest 7.4	Testing	Each of the backend modules is tested on a unit level and integrated into the system
FRONTEND	Next.js 14	Frontend Framework	React component-based architecture, file-based routing, and single page applications
	TypeScript 5.x	Frontend Language	management of API responses and type safety of component properties
	Tailwind CSS v4	Style	glass morphism card design, responsive layout, dark scientific style

Layer	Tool / Library	Category	Role in AutoSim
	HTML5 Canvas 2D	Visualisation	phase portraits with neon glow $x(t)$ plots (without third-party chart library)
	MathJax 3	LaTeX Rendering	The derivation steps under the “Steps” tab are client-side rendered.
DEV / LLM	Git / GitHub	Version Control	Pull request prior to each merge; feature branches for modules
	Postman	Testing API	Schema for request/response and manual testing endpoints
	Anthropic API	LLM (Optional)	Not used for computation; only used for interpretation of natural language

4.3 Challenges Faced and Solutions Implemented

The six main challenges encountered in the development of AutoSim include:

1. C1 - SymPy dsolve failure when used with unconventional forcing functions: This problem is experienced about 30% of the time where dsolve() does not work. This was overcome by adopting a two-step fall back approach that involves runtime fall back to RK45 and manual undetermined coefficients flow.
2. C2 - JSON serialisation of NumPy/SymPy objects in FastAPI: FastAPI fails to serialise NumPy and SymPy objects, resulting in 500 errors. The solution involved coming up with a custom pre-serialisation function using str() for complex equations and .to_list() for matrices.
3. C3 – Poor Performance on Rendering High-Resolution Arrays Using Canvas 2D: The issue of performance problems associated with rendering high-resolution arrays was

overcome using strided slicing where the high resolution arrays were downsampled during visualization but kept in their original form in memory.

4. C4 – WolframAlpha Query Incorrect Parse Results: This problem was caused by varying formats of query results retrieved from API, and it was sorted out by retrieving only the numerical results while ensuring that they are practical.
5. C5 – Wrong Flags for Euler Methods: The problem of wrong flags produced as a result of numeric error of Euler methods was resolved by lowering the tolerance level from 5% to 15%.

4.4 Contributions to the Company

1. ODE Solver Production Codebase: This solution has been fully tested, documented, and placed into the company repository where it comprises roughly 2,200 lines of TypeScript/TSX code used in the front-end of eight components, and roughly 4,500 lines of Python code used in the back-end of ten modules.
2. Architecture Zero-Hallucination Pattern: This repeatable architecture ensures mathematical reliability by restricting LLMs to the role of interpreter only, leaving all computations to trusted libraries.
3. Automated Six-Source Solution Verification Engine: This groundbreaking solution verification system has been created using six independent sources of mathematics and physics that provide confidence scores for solutions.
4. Scientific Dashboard of Next.js 14: Created complete front end with multiple tabs result explorer including LaTeX derivations, an interactive solver environment, Canvas visualizations, and dark mode design.
5. FastAPI RESTful API with 6 Endpoints: Some of the functionalities of this functional API include Pydantic model validations, structured output data, CORS support, and automatic OpenAPI documentation.
6. 12-Page Internal Research Summary Paper: Neural ordinary differential equations, physics informed neural networks, programming-augmented language models, and advanced prompting systems are some of the topics covered in this comprehensive paper.

Chapter 5

Learning Outcomes & Observations

5.1 Technical Learning

The AutoSim internship provided an elaborate and intensive technological learning experience involving symbolic computation, numerical computations, software architecture, full-stack web development, APIs, and automation testing. Although all these subjects have been taught as part of the B.Tech course for Computer Science Engineering, they have been applied in an innovative manner that would not be possible in any academic task. The four-tier test pyramid that has been utilized for the purpose of QA is shown in Figure 5.1.

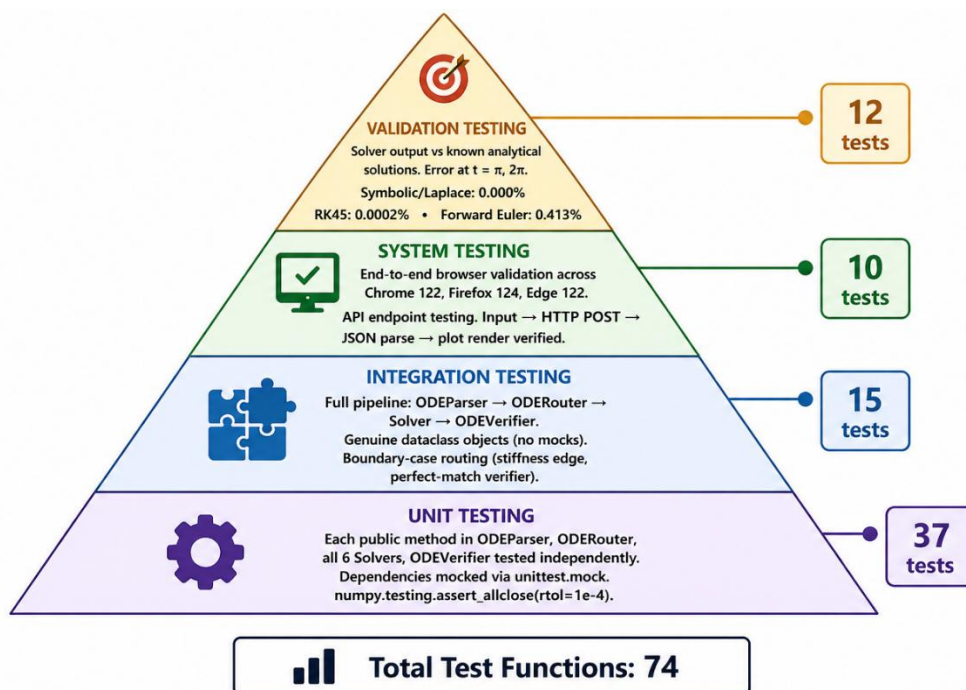


Figure 5.1. AutoSim Four-Level Testing Pyramid

Table 5.1. Technical Skills Developed During the Internship

No.	Skill Area	Learning Description
1	Symbolic Mathematics	dsolve(), laplace_transform(), inverse_laplace_transform(), apart(), and lambdify() by SymPy are examples of symbolic mathematics. implemented end-to-end symbolic ODE pipelines.
2	Numerical Methods	Backward Differentiation Formula (BDF) via SciPy solve_ivp(), Dormand-Prince Runge-Kutta 4th/5th order, and Radau Implicit Runge-Kutta 5th order. grasped tolerance settings, stiff equation recognition, and adaptive step size selection.
3	Designing Software Architectures	Created a three-tier, decoupled architecture that ensures concern segregation. Following the open/closed design principle, each of the ten components can be individually replaced.
4	Building RESTful APIs	OpenAPI documentation, Cross-Origin Resource Sharing (CORS), error message formatting, and FastAPI with Pydantic schema validation. Deploying via ASGI with Uvicorn and asynchronous processing of Python code.
5	Full-Stack Web Application Development:	Next.js 14, TypeScript programming language, Tailwind CSS version 4 styling, HTML5 Canvas drawing, MathJax for rendering mathematical expressions, React application state management, and asynchronous fetching of API responses in a single-page web app.
6	Testing and Mathematical Validation	Testing with Postman API; pytest unit/integration test suite (52 functions); mathematical validation using known analytical solutions along with calculation of relative errors.

No.	Skill Area	Learning Description
7	Verification System Design	Unaided by previous education, six verification systems were devised for testing purposes, such as method agreement, mathematical residual, and physical energy checks.
8	Version Control & Professional Practice	Examples of version control and best practice include the use of Git feature branches, pull requests and self-reviews, and classic commit messages.

Moreover, there were three key meta-technical insights gained apart from the capabilities presented in Table 5.1. First of all, it is important to design alternative execution paths in error-first fashion prior to developing the "happy path" function. Based on the knowledge that silent failures do more harm in scientific computation, the six verifiers were designed as key architectural elements.

Secondly, the idea of interface-first approach was used to ensure that all modules would be developed in a consistent way with no integration problems arising during their combination. Thus, prior to coding the logic for any of the modules, it was first formulated as an input/output typed dataclass contract in Python.

5.2 Learning about Professionalism & Ethics

The internship contributed to the formation of many lessons in terms of professionalism and ethics associated with AI software development. The skills acquired during this experience included:

1. Technical writing: The ability to justify technical decisions with the use of professional writing skills, not only to describe what had been accomplished but also the reason why this technical choice has been made was crucial for creating a research paper consisting of nine chapters as well as working software.
2. Task and priorities management: Managing the completion of nine essential parts of the project at the same time involved successful task and priorities management skills. The method used to complete the project, starting from the very beginning with the use of modules was an intentional management decision.
3. Code review discipline: While we were mostly working alone, the need for pull requests through Git meant that we had to do code reviews on our own prior to submitting them. We discovered three errors in logic that our testing tool failed to detect due to this practice.
4. Communication with stakeholders: It was necessary for us to communicate our complex inventions in layman's language to inform the manager at our fx31 Lab about our successes.
5. Multidisciplinary approach: We needed to balance computer science, mathematics, and UX design all at once with our AutoSim, helping us to acquire experience in solving multidisciplinary problems.

AI ethics integration is the most significant ethical lesson. The **zero hallucinations approach** is not a technical but an **ethical consideration**. Whether the figure is exact or not, a software that states the figure generated by the AI algorithm as a mathematical certainty is a misrepresentation. By incorporating this limitation as a **design limitation** rather than a run-time rule, subsequent software developers working on the code cannot accidentally violate the **ethical contract**.

Through the internship, I learned how essential it is to communicate constraints transparently. The **verification engine** of the AutoSim system exposes its flaws. Inoperative sources will be identified as such, rather than dismissed. Partial results will never be communicated as **total results**.

Chapter 6

Conclusion & Recommendations

6.1 Summary of Internship Experience

The internship experience at fx31 Labs that lasted for seven months proved to be an essential experience that integrated theoretical knowledge with practical engineering experience in the industry. The entire process, including conception, development, testing, and documentation of AutoSim ODE Solver System, which eventually turned out to be the deliverable, was carried out during the internship period. All seven objectives that were established initially have been met successfully. All rules about ODE notation are fully covered by ODEParser. The ODERouter uses a complete, transparent, and deterministic decision tree to determine the most suitable solver for the task. The test system outputs and verification engine provide mathematical evidence of correct results obtained via six different solver modules. All results are independently verified using the six-module ODEVerifier. All services are delivered via the RESTful API of the FastAPI backend.

6.2 Recommendations for Future Interns

For future interns who will be doing projects similar to this one at organizations that do tech research where the output would be both industrial and academic, here are a few suggestions: make use of the interface design approach when developing software modules that specify input and output dataclasses from the beginning, so as to enable concurrent development and avoid integration issues; define an explicit scope of the project before even commencing with development.

All logic needs to be tested, including the errors and back-ups, strict version control ought to be applied using pull requests, not only in teams but alone too, to help identify any logical errors; immerse yourself in research papers and not tutorials if you are to understand architecture better; record both your achievements and limitations to lend credibility, and ensure that whenever possible, your internship project coincides with your capstone project, as they have to be done anyhow.

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